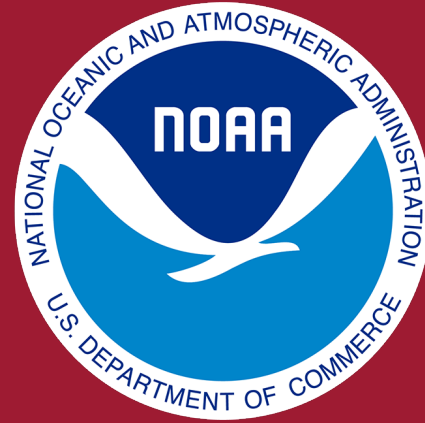


CIROH's Community Water Modeling Capabilities

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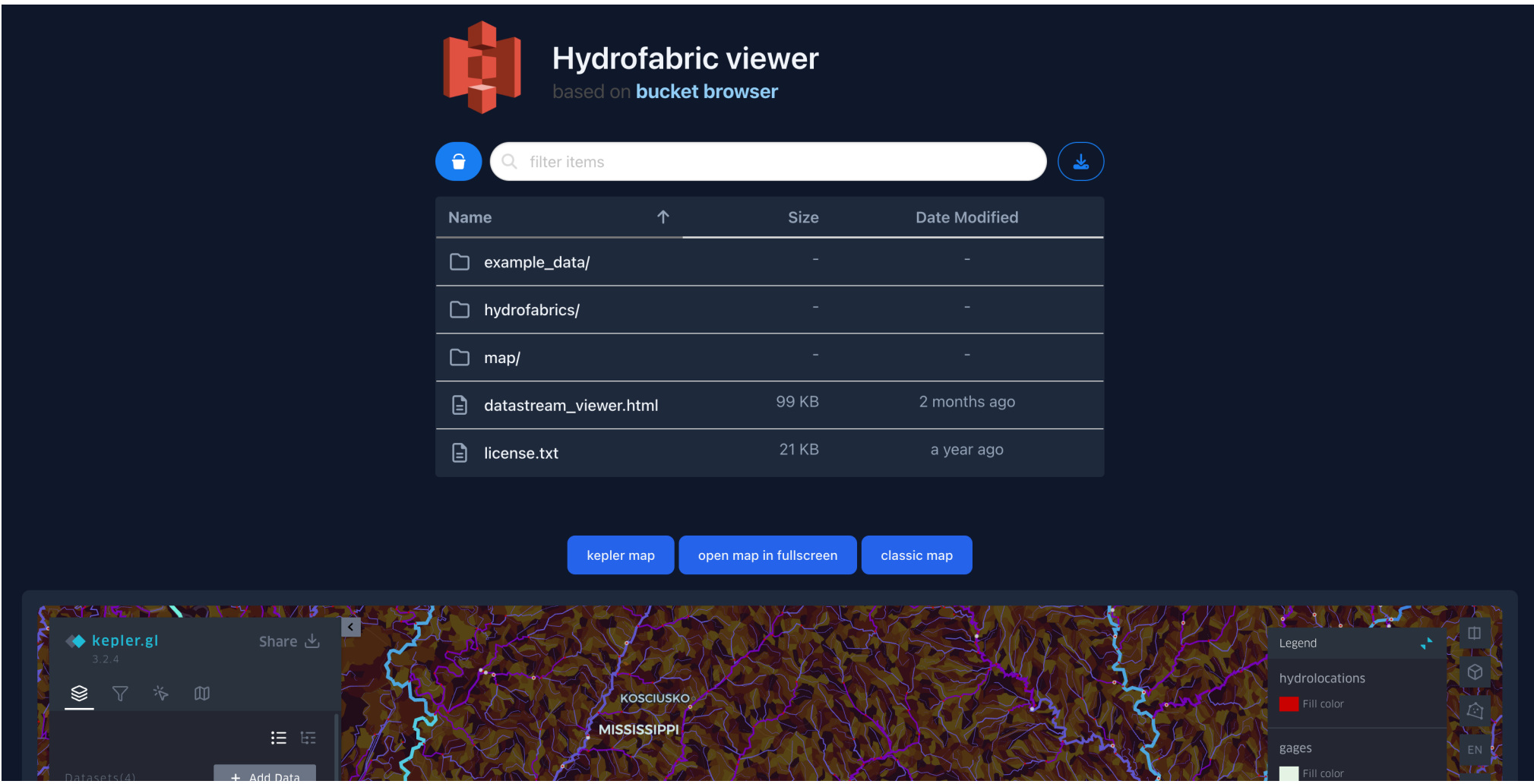
Digital Poster

For an interactive experience, including videos, links, demos, references, and acknowledgments:



Community Hydrofabric

- Data on **soil attributes, model parameters, and geometry**
- Community patches update hydrofabric parameters
- Contains map assets that power visualizers and apps



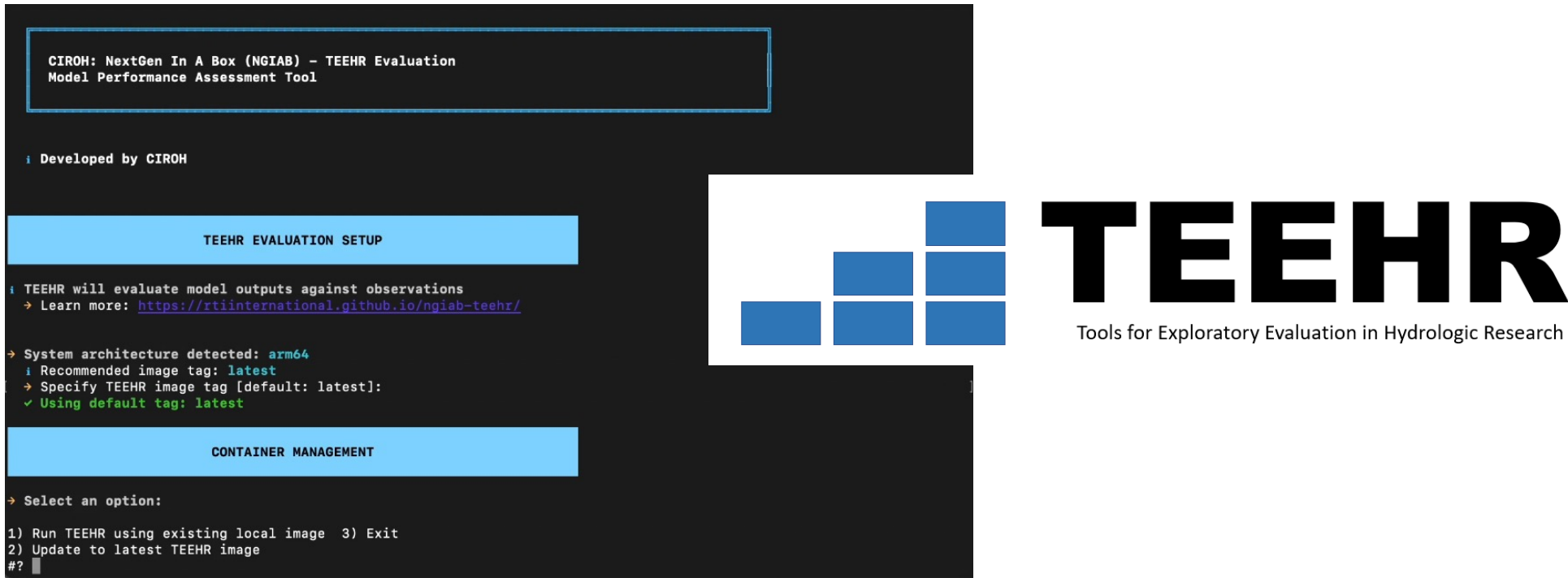
Data Preprocessor

- Python package, **prepares data for a NextGen-based simulation**
- Subsets hydrofabric, calculates forcings, creates configurations
 - Forcings from National Water Model Retrospective v3 or Analysis of Record for Calibration
 - Hydrofabric and configuration attributes from Community Hydrofabric
- Supports CFE/NOM, dHBV2, SUMMA, SNOW17, SAC-SMA



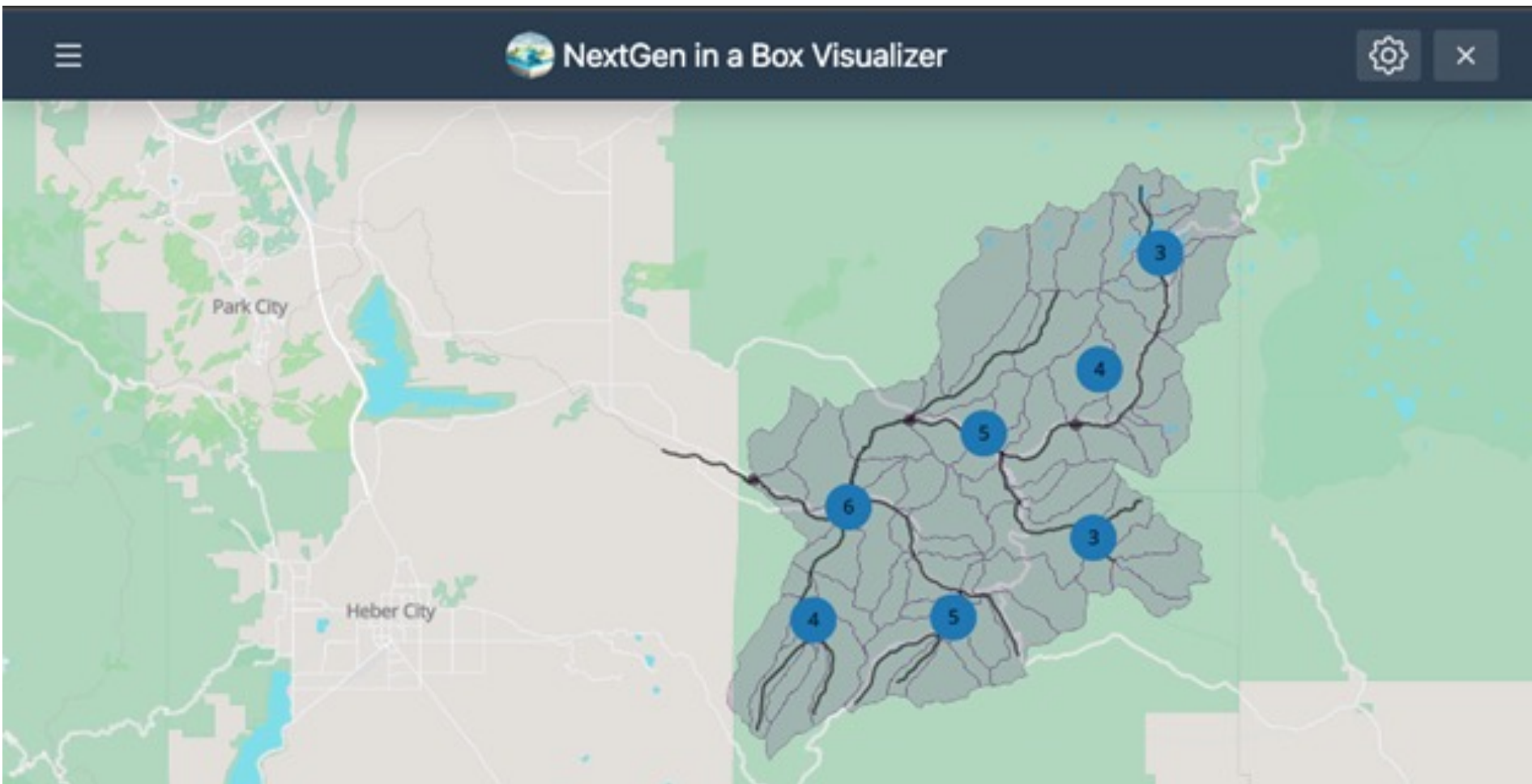
TEEHR

- Python package for **evaluation of hydrologic model performance**
- TEEHR coupled with NGIAB



NGIAB-NRDS Visualizer

- **Web-based exploration of model outputs**, built with Tethys and React
- Geospatial visualization of catchments and nexus points
- Time series analysis of outputs
- TEEHR output visualization
- Visualizer coupled with NGIAB



NextGen in a Box (NGIAB)

- Open-source **containerization of the NextGen framework**
- End-to-end workflow for developing and running NextGen-based models
- Reproducible across local, cloud, and high-performance computing environments

NextGen In A Box (NGIAB): Open-Source Containerization of the NextGen Framework to Enable Community-Driven Hydrology Modeling

Background	Implementation	NGIAB Ecosystem
The NextGen framework holds immense promise for connecting hydrological research with operational models... ...but setting up the framework is technically demanding and time-consuming	Containerization Reproducible environments across any platform Supportive tooling CI/CD Pipeline Automates release management & distribution	Community Hydrofabric Data Preprocessor NGIAB Model Execution TEEHR Integration Data Visualizer Community resources NGIAB 101

Patel et al, 2025 | Environmental Modeling & Software
Funded under award NANWS4320004 from NOAA CI Program
<https://ngiab.ciroh.org/>

NGIAB-Cal

- Python package for **calibrating parameters** for a NGIAB model run
- User-defined calibration configuration
- Calibrates the CFE and NOM models
- Single-gage calibration
- Uses Dynamically Dimensioned Search (DDS) algorithm

